

Lecture no 6

Dispersion, Variance and Standard Deviation

Dispersion ^{tells spread of data}

Mean: 16

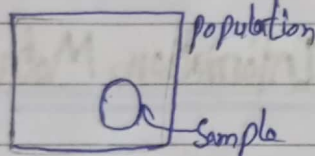
Median: 16

14 14 14 14 18 18 18 18

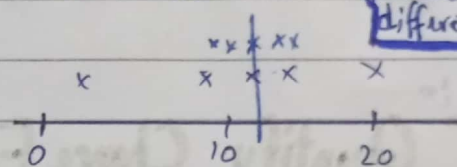
16 16 16 16 16 16 16 16

Mean: 16

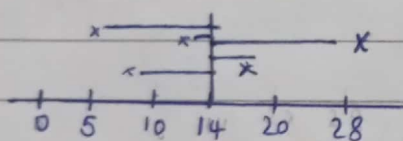
Median: 16



$$\text{Mean} = \bar{X} = \left(\sum_{i=1}^N x_i \right) / N$$



"Dispersion" = "Average of values from mean" ^{squared}



$$\begin{array}{l} x_i - \bar{x} \\ 28 - 14 = 14 \\ 5 - 14 = -9 \end{array}$$

$$\begin{array}{l} \downarrow \\ 14^2 = 196 \\ (-9)^2 = 81 \end{array}$$

The reason we not use absolute value is that it is not well defined at minimal value and we can't take derivative at that point

So, from all above we come to a point

where

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N}$$

"variance" $\downarrow$ "sigma" symbol

Q: why can't absolute value instead of $(x_i - \bar{x})^2$?

Problem: Original data units: m

Variance units: m^2

Solution: Square root!

$$\sigma = \sqrt{\sigma^2}$$

Standard Deviation: "Square root of average square distances of values from the mean"

"How far away are values from the mean ..."

All code of lecture 6 is done under file name
"03_dispersion.ipynb".

How large is SD?

Is 10.7 large?

How about 1094?

Ans - Depends on the mean.

$$\frac{10.7}{100} \times 100 = 10.7$$

$$\frac{1094}{100,000} \times 100 = 1.09$$

Coefficient of variation: $\frac{\sigma}{\bar{x}} * 100$

"Relative measure of deviation"